

Pressure Drop in Laminar Developing Flow in Noncircular Ducts: A Scaling and Modeling Approach

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A detailed review and analysis of the hydrodynamic characteristics of laminar developing and fully developed flows in noncircular ducts is presented. New models are proposed, which simplify the prediction of the friction factor–Reynolds product $f Re$ for developing and fully developed flows in most noncircular duct geometries found in heat exchanger applications. By means of scaling analysis it is shown that complete problem may be easily analyzed by combining the asymptotic results for the short and long ducts. Through the introduction of a new characteristic length scale, the square root of cross-sectional area, the effect of duct shape has been minimized. The new model has an accuracy of $\pm 10\%$ or better for most common duct shapes when nominal aspect ratios are used, and $\pm 3\%$ or better when effective aspect ratios are used. Both singly and doubly connected ducts are considered. [DOI: 10.1115/1.4000377]

Keywords: laminar flow, noncircular ducts, hydrodynamic entrance length, developing flow, modeling, pressure drop

1 Introduction

Laminar flow fluid friction and heat transfer in noncircular ducts occur quite frequently in low Reynolds number flow heat exchangers such as automotive coolers, cold plates, and micro-channel heat sinks. It is now also occurring more frequently in a host of other energy conversion and reclamation devices, as a result of the miniaturization of modern technologies. While traditional approaches have relied heavily on the use of tabulated and/or graphical data, the ability to design thermal systems using robust models is much more desirable in the age of computer simulations and computer assisted design. Most modern fluid dynamics and heat transfer texts rarely present correlations or models for more complex geometries, which appear in many engineering systems. Rather, a subset of data for miscellaneous geometries is usually presented after detailed discussion and analysis of simple geometries such as the circular duct and parallel plate channel.

In the present paper, the hydrodynamic problem is considered in detail, and a new and much simpler model is developed for predicting the friction factor–Reynolds number product for developing laminar flow in noncircular ducts.

Laminar fully developed fluid flow in noncircular ducts of constant cross-sectional area results when the duct length L is sufficiently greater than the hydrodynamic entrance length L_h , i.e., $L \gg L_h$, or when the characteristic transversal scale is sufficiently small to ensure a very small Reynolds number. Under these conditions the flow through most of the duct or channel may be considered fully developed. However, in many engineering systems such as compact heat exchangers and microcoolers used in electronics packaging, while the characteristic dimension of the flow channel is small enough to give rise to laminar flow conditions, the flow length is generally not sufficiently large enough to give rise to fully developed flow, i.e., $L \sim L_h$ or $L \ll L_h$, and developing

flow prevails over most of the duct length. In these situations, a model capable of predicting the hydrodynamic characteristic, usually denoted as $f Re$, the friction factor–Reynolds number product as a function of dimensionless duct length, is required.

Since the pressure drop in a developing flow is due to both wall shear and fluid acceleration, some references choose to denote it as the *apparent* friction factor–Reynolds number product $f_{app} Re$, in order to distinguish it from the fully developed flow value $f Re$. However, since dimensionless pressure drop or $f_{app} Re$ in the entrance region transitions smoothly to $f Re$ in fully developed flow, this distinction is dropped in favor of just $f Re$, which is implied to vary with dimensionless duct length, i.e., $f Re(L^+)$, where L^+ is the dimensionless duct length to be defined shortly.

2 Literature Review

A review of literature reveals that only two significant attempts at developing a general model have been undertaken. These are the work of Shah [1] and Yilmaz [2]. Both of these models are based on the earlier work of Bender [3]. Bender [3] combined the asymptotic result for a “short” duct [4], with the result for the “long” duct, to provide a model that is valid over the entire length of a circular duct. The demarcation between short and long being the duct length relative to the hydrodynamic entrance length as shown in Fig. 1, i.e.,

$$f Re_{D_h} = \begin{cases} \frac{3.44}{\sqrt{L^+}}, & L^+ < 0.001 \\ (f Re)_{fd} + \frac{K_\infty}{4L^+}, & L^+ > 0.1 \end{cases} \quad (1)$$

where

$$L^+ = \frac{L/D_h}{Re_{D_h}} \quad (2)$$

is the dimensionless duct length.

As such, this formulation requires the use of the concept of the incremental pressure drop factor K_∞ . Shah [1] later extended the model of Bender [3] to predict results for the equilateral triangle,

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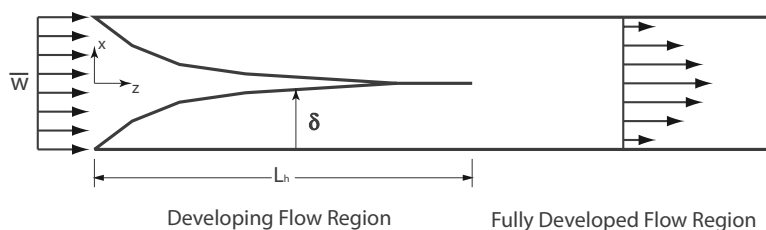


Fig. 1 Hydrodynamic entrance problem

the circular annulus, the rectangular duct, and parallel plate channel geometries. Shah [1] achieved this by generalizing the form of the model of Bender [3], and by tabulating coefficients for each particular geometry for a number of channel and annulus aspect ratios using

$$f Re_{D_h} = \frac{3.44}{\sqrt{L^+}} + \frac{(f Re)_{fd} + K_{\infty}/4L^+ - 3.44/\sqrt{L^+}}{1 + C/(L^+)^2} \quad (3)$$

Yilmaz [2] proposed a more general model of Shah [1]. Rather than tabulating coefficients, Yilmaz [2] developed a complex correlation scheme for the fully developed friction factor $(f Re)_{fd}$, the incremental pressure drop K_{∞} , and a fitting coefficient C , which appears in the Shah [1] model. This model is more general than that of Shah [1] but is also quite complex since a base correlation has been augmented with three additional correlations.

Despite its complexity, the model of Yilmaz [2] is accurate over the entire range of the entrance and fully developed regions for many duct cross sections. The primary drawback of the simple model proposed by Shah [1] is the requirement of tabulated coefficients and parameters for each geometry, i.e., $(f Re)_{fd}$, K_{∞} , and C , thus limiting interpolation for geometries such as the rectangular duct and circular annulus, whose solution varies with aspect ratio. In the case of the model developed by Yilmaz [2], interpolation is no longer a problem; however, this is achieved at the cost of simplicity.

The two models discussed above represent the current state of the art for internal flow problems. Both models are based on the combination of the short duct and long duct solutions using the correlating method of Bender [3]. In this approach the incremental pressure drop factor K_{∞} is required in the long duct solution (see Eq. (1)). As a result of the complex correlating equations for K_{∞} developed by Yilmaz [2], the simple physical behavior of the hydrodynamic entrance problem is lost. In both cases, however, the models are based on the proposed form put forth by Bender [3]. Unfortunately, this form of correlation is more complicated than need be for this problem. It was formulated as a nonlinear superposition of two asymptotes. However, only one of these asymptotes is a true asymptote. The other asymptote represents a transitional one, which contains the true fully developed asymptote. As such, the model introduces the incremental pressure drop, which is not actually required. It is apparent from the available data that smooth transition occurs from the entrance region to that of fully developed flow.

Since the solution obtained by Shapiro et al. [4] accounts for the increase in momentum of the accelerating core and the wall shear, the use of the term K_{∞} in a hydrodynamic entrance model such as that proposed by Bender [3] is redundant. The model presented in this paper does not require this parameter and is significantly simpler in composition.

3 Governing Equations

The governing equations for steady incompressible flow in the hydrodynamic entrance region in a noncircular duct or channel are

$$\nabla \cdot \vec{V} = 0 \quad (4)$$

and

$$\rho \vec{V} \cdot \nabla \vec{V} = -\nabla p + \mu \nabla^2 \vec{V} \quad (5)$$

Simultaneous solution to the continuity, Eq. (4), and momentum, Eq. (5), equations subject to the no slip condition at the duct wall, $\vec{V}=0$, the boundedness condition along the duct axis, $\vec{V} \neq \infty$, and a constant initial velocity, $\vec{V}=U\vec{k}$, is required to characterize the flow. However, most available data are a result of solution to a much simpler set of equations, which result from boundary layer type assumptions [4].

In the case of cross sections that can be characterized by a single variable, these yield the continuity equation and momentum equation in the direction of the flow, i.e., the z -momentum equation. For two dimensional cross sections, an additional closure equation is required for relating transverse velocity components. Additional discussion on special issues pertaining to the solution of this problem is discussed in Ref. [4].

In Sec. 4, scaling analysis is used to show the appropriate form of the solution for both short and long ducts. Later, asymptotic analysis is used to develop a new model. Several of the duct shapes of interest are shown in Fig. 2.

Solutions to Eqs. (4) and (5) have been obtained using a host of methods including finite difference methods, variational methods, integral methods, and theoretical approaches, which ultimately require some form of numerical method. A simple modeling approach is presented, which predicts that all of the available data are now developed.

4 Scale Analysis

We now examine the momentum equation and consider the various force balances implied under particular flow conditions. The momentum equation represents a balance of three forces: inertia, pressure, and friction, i.e.,

$$\underbrace{\rho \vec{V} \cdot \nabla \vec{V}}_{\text{inertia}} = \underbrace{-\nabla p}_{\text{pressure}} + \underbrace{\mu \nabla^2 \vec{V}}_{\text{friction}} \quad (6)$$

Geometry	Shape	P/A
Rectangle		$\frac{2(\epsilon + 1)}{\sqrt{\epsilon}}$
Ellipse		$\frac{4E(\epsilon')}{\sqrt{\pi\epsilon}}$
Annulus		$\frac{2\sqrt{\pi}(1+r^*)}{\sqrt{(1-r^{*2})}}$
Polygons		$2\sqrt{N} \left(\tanh\left(\frac{\pi}{N}\right) \right)^{1/2}$
Isosceles Triangle		$\frac{2/\cos\phi + 2\tan\phi}{\sqrt{\tan\phi}}$

Fig. 2 Common duct shapes

We now consider three separate force balances. Each is examined below using the method of scale analysis advocated by Bejan [5]. Using Fig. 1 as a starting point, we will examine the various force balances implied by Eq. (5) in the two regions divided by L_h .

4.1 Long Duct Asymptote, $L \gg L_h$. Fully developed laminar flow in a duct of arbitrary, but constant cross section, is governed by the Poisson equation

$$\underbrace{\nabla p}_{\text{pressure}} = \underbrace{\mu \nabla^2 \vec{V}}_{\text{friction}} \quad (7)$$

which represents a balance between the friction and pressure forces. Thus we may write the following approximate relation using the characteristics of the flow and the geometry:

$$\frac{\Delta p}{L} \sim \mu \frac{U}{\mathcal{L}^2} \quad (8)$$

where $\mathcal{L}$ represents a characteristic transversal length scale of the duct cross section. The velocity scales according to the area mean value U , and the axial length scales according to L . Rearranging the above expression gives

$$\Delta p \sim \frac{\mu UL}{\mathcal{L}^2} \quad (9)$$

Next, we examine the shear stress at the wall. The shear stress may be approximated by

$$\vec{\tau} = \mu \nabla \vec{V} \sim \mu \frac{U}{\mathcal{L}} \quad (10)$$

Next, we may introduce the definition of the friction factor, defined as the dimensionless wall shear

$$f \sim \frac{\tau}{\rho U^2} \sim \frac{\mu U / \mathcal{L}}{\rho U^2} \sim \frac{1}{\text{Re}_{\mathcal{L}}} \quad (11)$$

The above expression may be written such that the following relationship exists for all cross-sectional geometries:

$$f \text{Re}_{\mathcal{L}} \sim \frac{\tau \mathcal{L}}{\mu U} \sim \mathcal{O}(1) \quad (12)$$

or

$$f \text{Re}_{\mathcal{L}} = C_1 \quad (13)$$

Equation (12) is often denoted as the Poiseuille number Po in modern literature, such that $f \text{Re}_{\mathcal{L}} = 2 \text{Po}_{\mathcal{L}}$. The factor of 2 results from the use of the kinetic or dynamic pressure in Eq. (11), i.e., $\frac{1}{2} \rho U^2$, which is omitted in the scaling analysis. The constant C_1 has been found to vary for most geometries in the range $12 < C_1 < 24$, when $\mathcal{L} = D_h$.

Finally, a control volume force balance gives

$$\tau = \frac{\Delta p A}{L P} \quad (14)$$

and yields the following relationship when combined with the two scaling laws, Eqs. (10) and (14):

$$\mathcal{L} \sim \frac{A}{P} \sim \frac{D_h}{4} \quad (15)$$

We shall see later that this length scale, which results from the force balance, although convenient, is not the most appropriate choice. As a length scale it is fundamentally important in the relationship of Eq. (14), but this by no means implies necessity of its use in Eq. (12). Equation (12) represents a dimensionless mean wall shear, which gives us freedom to choose $\mathcal{L}$. It is in this definition that the appropriate choice of a characteristic length scale must be carefully considered.

4.2 Short Duct Asymptote, $L \ll L_h$. In the entrance region near the duct inlet two regions must be considered. One is the inviscid core and the other the viscous boundary layer. The flow within the boundary layer is very similar to laminar boundary layer development over a flat plate, except that the velocity at the edge of the boundary layer is no longer constant in the duct flow. The two regions are now examined beginning with the flow within the boundary layer.

The force balance within the boundary layer is governed strictly by inertia and friction forces

$$\underbrace{\rho \vec{V} \cdot \nabla \vec{V}}_{\text{inertia}} = \underbrace{\mu \nabla^2 \vec{V}}_{\text{friction}} \quad (16)$$

This balance yields

$$\frac{U^2}{L} \sim \mu \frac{U}{\delta^2} \quad (17)$$

where δ is the boundary layer thickness (see Fig. 1). Thus,

$$\frac{\delta}{L} \sim \frac{1}{\sqrt{\text{Re}_L}} \quad (18)$$

Although the flow in a confined channel is markedly different from that of a plate in streaming flow, both experience boundary layer growth, which is inversely proportional to the Reynolds number. The major distinction, as we shall see is the rate at which the boundary layer grows within a confined channel, is much slower than a plate in streaming flow, due to a favorable pressure gradient. Introducing the transversal length scale $\mathcal{L}$ we obtain

$$\frac{\delta}{\mathcal{L}} \sim \sqrt{\frac{L}{\mathcal{L} \text{Re}_{\mathcal{L}}}} \sim \sqrt{L^+} \quad (19)$$

where L^+ is the dimensionless duct length defined by

$$L^+ = \frac{L}{\mathcal{L} \text{Re}_{\mathcal{L}}} \quad (20)$$

Next, considering the relationship for the wall shear

$$\vec{\tau} = \mu \nabla \vec{V} \sim \mu \frac{U}{\delta} \quad (21)$$

and the friction factor, we obtain

$$f \sim \frac{\tau}{\rho U^2} \sim \frac{\mu U / \delta}{\rho U^2} \sim \frac{\mu \sqrt{\text{Re}_L}}{\rho U L} \sim \frac{1}{\sqrt{\text{Re}_L}} \quad (22)$$

Rewriting the above expression in terms of the length scale $\mathcal{L}$, and defining the product of friction factor and Reynolds number yield the following expression for the entrance region:

$$f \text{Re}_{\mathcal{L}} \sim \frac{\mathcal{O}(1)}{\sqrt{L^+}} \quad (23)$$

or

$$f \text{Re}_{\mathcal{L}} = \frac{C_2}{\sqrt{L^+}} \quad (24)$$

Equation (19) illustrates an important point highlighted by Bejan [5], that hydrodynamically developing flows should be presented as a function of $\sqrt{L^+}$ rather than L^+ , since the effective wall shear is inversely proportional to δ and hence a function of $\sqrt{L^+}$ as shown in Eq. (24).

Finally, in the inviscid core, the momentum equation represents a balance of inertia and pressure forces

$$\underbrace{\rho \vec{V} \cdot \nabla \vec{V}}_{\text{inertia}} = \underbrace{-\nabla p}_{\text{pressure}} \quad (25)$$

which scales according to

$$\rho U^2 \sim \Delta p \quad (26)$$

Since the boundary layer in the developing region will be thinner under a favorable pressure gradient in the core, the constant in Eq. (24) will be much larger than the value obtained for friction in a boundary layer flow over an isolated flat plate in streaming flow. The constant C_2 has been found theoretically by Siegel [6] for the circular duct to be $C_2=3.44$ for a mean friction factor and $C_2=1.72$ for a local friction factor. For an isolated plate in streaming flow, these are $C_2=1.328$ and $C_2=0.664$, respectively. As will be seen shortly, the boundary layer develops over a length four times greater than that predicted by the simple Blasius theory. Thus it is reasonable to expect that the friction factor would be greater by approximately the same value, given the inverse dependency of f on δ . In this case it is approximately 2.6 times greater.

In summary, we have found from scaling analysis the following relationships for the friction factor–Reynolds number product:

$$f \text{Re}_{\mathcal{L}} = \begin{cases} C_1, & L \gg L_h \\ \frac{C_2}{\sqrt{L^+}}, & L \ll L_h \end{cases} \quad (27)$$

This asymptotic behavior will be examined further and will form the basis for the new model developed for the hydrodynamic entrance problem, once the exact limits are formulated.

4.3 Hydrodynamic Entrance Lengths. The entrance length is traditionally defined as the point downstream of the inlet, where the centerline velocity is 99% of the maximum velocity. This measure of the development region is also utilized in the selection of an appropriate heat transfer model as well as ascertaining which type of friction model be utilized for pressure drop calculations. One may easily get an estimate for the entrance length using simple boundary layer theory.

If the entrance length is used as a criterion to measure the extent of the boundary layer region, then we shall refer to a *short duct* and *long duct* in the following manner:

$$\begin{aligned} L &\ll L_h && \text{short duct} \\ L &\gg L_h && \text{long duct} \end{aligned} \quad (28)$$

We use the much greater and much less than notation, since a channel flow, which only marginally satisfies the entrance length criteria, i.e., $L \sim L_h$, experiences a significantly higher pressure drop than a fully developed flow. Care must be taken to ensure that fully developed flow prevails over most of a duct or channel. Merely stating that $L > L_h$ is not strictly valid for making an assumption of fully developed flow.

If we consider a simple channel formed by two plates separated by $2b$, the boundary layers on each plate in laminar flow will merge at a point $z \sim L_h$ downstream, when their thickness is of order of the half plate separation

$$\delta = \frac{5z}{\sqrt{\text{Re}_z}} \sim b \quad (29)$$

or

$$L_h \sim 0.0025 D_h \text{Re}_{D_h} \quad (30)$$

after introducing $D_h=4b$ for the channel. This estimate is much shorter than the actual length in a channel flow, as the favorable pressure gradient in the inviscid core leads to much slower boundary layer development than the case for Blasius flow. The actual solution for a channel is given by [4]

$$L_h \sim 0.011 D_h \text{Re}_{D_h} \quad (31)$$

which is approximately four times greater. Thus, given that the boundary layer develops more slowly in a channel flow, we easily see from Eqs. (19), (21), and (24) that the constant $C_2 > 1.328$, owing to the acceleration within the inviscid core.

Depending on the channel or duct shape, the entrance length may be written more generally as

$$L_h = C_{0,D_h} D_h \text{Re}_{D_h} \quad (32)$$

where C_{0,D_h} varies weakly with duct shape. It typically has an order of magnitude $0.01 < C_{0,D_h} < 0.06$, when $\mathcal{L}=D_h$.

A simple equation relating the approximate magnitude of the hydrodynamic entrance length may be obtained using scaling analysis, by considering an equality between the two asymptotic limits given in Eq. (27) when $L^+=L_h^+$,

$$C_1 = \frac{C_2}{\sqrt{L_h^+}} \quad (33)$$

or

$$L_h^+ = \left(\frac{C_2}{C_1} \right)^2 \sim C_{0,\mathcal{L}} \quad (34)$$

Later, it will be shown that this approximate scaling result compares well with more exact solutions. This solution represents the intersection of the asymptotic limits on a plot of the complete behavior of $f \text{Re}$ versus L^+ , whereas the traditional entrance length is defined on the basis of the centerline velocity.

4.4 Length Scales and Slenderness. One issue that has not been adequately addressed in literature is the selection of an appropriate characteristic length scale, i.e., $\mathcal{L}$. Traditionally, the hydraulic diameter has been chosen, $\mathcal{L}=4A/P$. However, in many texts, its use in laminar flow has been questioned [7–9]. In an earlier work [10], the authors addressed this issue using dimensional analysis. It was determined that the widely used concept of the hydraulic diameter was inappropriate for laminar flow and the authors proposed using $\mathcal{L}=\sqrt{A}$ as a characteristic length scale, by considering other problems in mathematical physics for which the Poisson equation applies. A more detailed discussion and analysis on the use of $\mathcal{L}=\sqrt{A}$ may be found in Refs. [11,12]. More recently, Duan and Muzychka [13] and Muzychka and Edge [14] showed that the square root of flow area is also more appropriate for nondimensionalizing gaseous slip flows and non-Newtonian flows.

The choice of an appropriate characteristic length, which minimizes the effect of duct shape on the numerical value of the dimensionless friction factor–Reynolds number product, defined by Eq. (13), will now be examined. Given an arbitrary duct of cross-sectional area A , perimeter P , and length L , one can choose several combinations of these generic characteristics to obtain a length scale

$$\mathcal{L} = \frac{4A}{P}, \sqrt{A}, P, L, \sqrt{PL}, (AL)^{1/3} \quad (35)$$

in addition to a host of other possibilities defined using the particular duct or channel dimensions, i.e., semi-axis lengths, etc. In the heat transfer and fluid flow literature the convention is to use the hydraulic diameter $4A/P$ or hydraulic radius $2A/P$. This characteristic length arises naturally from a simple control volume balance on an arbitrarily shaped straight duct, i.e., Eq. (15).

The use of the hydraulic diameter is often considered the most convenient as a result of the relationship defined by Eq. (15). However, when the hydraulic diameter is used, the dimensionless grouping $f \text{Re}_{D_h}$ takes on values that vary between 12 and 24 for most common duct shapes. This requires the use of exact values for accurate prediction of the pressure drop. The basis for the use of the hydraulic diameter arose in the modeling of turbulent flows in noncircular ducts. In turbulent flows, the velocity gradient varies only in a region very near the wall and there is deeper penetration of the flow field into re-entrant corners. In a laminar flow the velocity gradient varies over most of the cross-sectional area and flow does not penetrate as deep into re-entrant corners. The result of these two effects is that a mismatch occurs between the true hydraulic equivalent duct and the hydraulic equivalent duct determined by the definition $4A/P$. However, a number of other

deficiencies in the hydraulic diameter concept should be addressed, namely, that the hydraulically equivalent circular area and perimeter based on the hydraulic diameter are not the same as the true area and perimeter of the noncircular duct. This mismatch in area and perimeter is often cited as the probable cause in the mismatch of dimensionless laminar flow data [7].

Other possibilities for $\mathcal{L}$ given above have a number of potential flaws. First, the perimeter and area are not definable for the parallel plate channel. This issue does not pose a problem for singly or doubly connected regions having finite area and perimeter. However, the parallel plate is merely an idealization for low aspect ratio ducts and channels. In principle it exists as a convenient limit; in practice, it is never truly realized. However, if one considers a duct of fixed area, the parallel plate channel is approached as the duct aspect ratio varied such that $a \gg b$, thus physically realizing the situation where $\sqrt{A}$ is a constant, but the duct aspect ratio $\epsilon \rightarrow 0$.

The use of perimeter may be considered from the point of view of constant shear stress or preservation of the retarding force and hence the surface area of the duct or channel. That is

$$(\bar{\tau}_w PL)_{nc} = (\bar{\tau}_w PL)_c \quad (36)$$

If the mean shear stress at the wall is constant, then the following relationship holds:

$$P_{nc} = P_c \quad (37)$$

and the effective circular diameter becomes

$$D_{eff} = \frac{P_{nc}}{\pi} \quad (38)$$

In other words, the characteristic length $\mathcal{L} = P \propto D_{eff}$.

Finally, it may also be argued on physical grounds that the square root of the flow area is essentially the same as preserving the cross-sectional duct area or maintaining a constant mass flux. That is

$$(\rho \bar{w} A)_c = (\rho \bar{w} A)_{nc} \quad (39)$$

Now if both the circular and noncircular ducts have the same mass flux $G = \rho \bar{w}$ then

$$A_c = A_{nc} \quad (40)$$

and the effective circular diameter becomes

$$D_{eff} = \sqrt{\frac{4A_{nc}}{\pi}} \quad (41)$$

In other words, the characteristic length $\mathcal{L} = \sqrt{A} \propto D_{eff}$.

It has been shown in Ref. [11] that $\mathcal{L} = \sqrt{A}$ succeeds in bringing the dimensionless results closer together for similar ducts, i.e., rectangular and elliptical or polygonal [10–14]. In all cases, better correlation of the laminar flow data was achieved versus an appropriate measure of duct slenderness, allowing a single approximate expression to be used for many duct shapes.

Finally, since it has been established that $\mathcal{L} = \sqrt{A}$ is more appropriate than the hydraulic diameter, we should consider its effect on the dimensionless duct length L^+ . Using Eq. (20) we may write either

$$L^+ = \frac{L/D_h}{\text{Re}_{D_h}} = \frac{\mu L P^2}{16 \dot{m} A} \quad (42)$$

or

$$L^+ = \frac{L/\sqrt{A}}{\text{Re}_{\sqrt{A}}} = \frac{\mu L}{\dot{m}} \quad (43)$$

Clearly, Eq. (43) is more appropriate as it does not contain the transverse length scale. It is more fundamental than Eq. (42),

Table 1 Definitions of aspect ratio

Geometry	Aspect ratio
Regular polygons	$\epsilon \approx 1$
Rectangle and ellipse	$\epsilon \approx \frac{b}{a}$
Miscellaneous shapes	$\epsilon \approx \frac{\ell_{\text{short}}}{\ell_{\text{long}}}$
Trapezoid	$\epsilon \approx \begin{cases} \frac{2h}{b_1 + b_2}, & h < (b_1 + b_2)/2 \\ \frac{b_1 + b_2}{2h}, & h > (b_1 + b_2)/2 \end{cases}$
Isosceles triangle	$\epsilon \approx \begin{cases} h/b & h < b \\ b/h & h > b \end{cases}$
Annular sector	$\epsilon \approx \frac{1 - r^*}{(1 + r^*)\Phi}$
Circular annulus	$\epsilon \approx \frac{(1 - r^*)}{\pi(1 + r^*)}$
Eccentric annulus	$\epsilon \approx \frac{(1 + e^*)(1 - r^*)}{\pi(1 + r^*)}$

since it contains only the mass flow rate, the viscosity, and the duct length.

The measure of a duct's slenderness can be easily defined for most simple shapes. If we consider the rectangular and elliptical ducts, a simple measure of slenderness is the ratio of the minor and major axes. In the case of triangular ducts, a measure of slenderness may be defined as the ratio of base to height or height to base, i.e.,

$$\epsilon \approx \frac{\ell_{\text{short}}}{\ell_{\text{long}}} \quad (44)$$

such that a simple intrinsic value falling in the range $0 < \epsilon < 1$ is obtained. However, even this simple measure requires some additional attention for ducts with small re-entrant corners.

In the case of the concentric annular duct, a simple measure of its proportions is traditionally the ratio of inner and outer radii. This measure is not compatible with that of most other shapes we consider, as slenderness is a combination of transverse lengths scales, while the radii ratio is a combination of parallel length scales. This issue can be overcome by defining a simple measure of slenderness

$$\epsilon \approx \frac{1 - r^*}{\pi(1 + r^*)} \quad (45)$$

This definition results from preserving the wall to wall spacing and defining a transverse length scale based on a rectangular duct of equivalent area. The issue becomes more complicated if one or both of the bounding ducts are polygons. But can be overcome if one defines

$$\epsilon \approx \frac{1 - \sqrt{A_i/A_o}}{\pi(1 + \sqrt{A_i/A_o})} \quad (46)$$

where A_i and A_o are the areas of the inner and outer polygons. The above formula is merely a generalized form where $\sqrt{A_i/A_o} = r^*$ for the concentric annulus.

A summary of the nominal aspect ratios of typical ducts is given in Table 1.

5 Asymptotic Analysis

In this section, asymptotic analysis [15] is used to establish expressions for the characteristic long duct and short duct behaviors established through scaling analysis. First, the long duct limit is considered and a simple expression developed for predicting the constant $f Re = C_1$. Additionally, the issue of an appropriate characteristic length scale is addressed and comparisons are made for both the hydraulic diameter D_h and square root of cross-sectional area $\sqrt{A}$ as characteristic length scales. Finally, the short duct limit is considered by re-examining the approximate solution obtained by Siegel [6] and Shapiro et al. [16].

5.1 Long Duct Asymptote, $L \gg L_h$. In order to establish the long duct limit, solutions for many ducts are examined. These solutions have been cataloged in Refs. [4,17]. The simplest duct shapes are the circular duct and the parallel plate channel. These important shapes also appear as limits in the elliptical duct, the rectangular duct, and the circular annulus.

We now examine a number of important results employing both $\mathcal{L} = 4A/P$ and $\mathcal{L} = \sqrt{A}$. Starting with the *elliptical duct*, the dimensionless average wall shear is found to be [4]

$$f Re_{D_h} = \frac{2\pi^2(1+\epsilon^2)}{E(\epsilon')^2} \quad (47)$$

where $\epsilon = b/a$, the ratio of minor and major axes, and $\epsilon' = \sqrt{1-\epsilon^2}$. Equation (47) has the following limits:

$$f Re_{D_h} = \begin{cases} 16, & \epsilon \rightarrow 1 \\ 19.76, & \epsilon \rightarrow 0 \end{cases} \quad (48)$$

If the solution is recast using $\mathcal{L} = \sqrt{A}$, as a characteristic length scale, the following relationship is obtained:

$$f Re_{\sqrt{A}} = \frac{2\pi^{3/2}(1+\epsilon^2)}{\sqrt{\epsilon}E(\epsilon')} \quad (49)$$

Equation (49) now has the following limits:

$$f Re_{\sqrt{A}} = \begin{cases} 8\sqrt{\pi} \approx 14.18, & \epsilon \rightarrow 1 \\ \frac{2\pi^{3/2}}{\sqrt{\epsilon}} \approx \frac{11.14}{\sqrt{\epsilon}}, & \epsilon \rightarrow 0 \end{cases} \quad (50)$$

Next, we examine the *rectangular duct*. The solution for the dimensionless average wall shear [4], considering only the first term of the series, gives

$$f Re_{D_h} = \frac{24}{(1+\epsilon)^2} \left[1 - \frac{192\epsilon}{\pi^5} \tanh\left(\frac{\pi}{2\epsilon}\right) \right] \quad (51)$$

Equation (51) has the following limits:

$$f Re_{D_h} = \begin{cases} 14.13, & \epsilon \rightarrow 1 \\ 24, & \epsilon \rightarrow 0 \end{cases} \quad (52)$$

Examination of the single term solution reveals that the greatest error occurs when $\epsilon = 1$, which gives a $f Re$ value 0.7% below the exact value of $f Re = 14.23$. The results for a wide range of aspect ratios are tabulated in Ref. [4] using a 30 term series, which provided 7 digit precision. A sample is provided in Table 2 for comparison with the simpler one term approximation.

If the solution is recast using $\mathcal{L} = \sqrt{A}$, as a characteristic length scale, the following expression is obtained:

$$f Re_{\sqrt{A}} = \frac{12}{\sqrt{\epsilon}(1+\epsilon)} \left[1 - \frac{192\epsilon}{\pi^5} \tanh\left(\frac{\pi}{2\epsilon}\right) \right] \quad (53)$$

Equation (53) now has the following limits:

Table 2 Comparison of single term approximation for $f Re$

$\epsilon = b/a$	$f Re_{D_h}$		$f Re_{\sqrt{A}}$		$12/\sqrt{\epsilon}$
	Exact	Equation (51)	Exact	Equation (53)	
0.001	23.97	23.97	379.33	379.33	379.47
0.01	23.68	23.68	119.56	119.56	120.00
0.05	22.48	22.47	52.77	52.77	53.66
0.1	21.17	21.16	36.82	36.81	37.95
0.2	19.07	19.06	25.59	25.57	26.83
0.3	17.51	17.49	20.78	20.76	21.91
0.4	16.37	16.34	18.12	18.09	18.97
0.5	15.55	15.51	16.49	16.46	16.97
0.6	14.98	14.94	15.47	15.43	15.49
0.7	14.61	14.55	14.84	14.79	14.34
0.8	14.38	14.31	14.47	14.40	13.42
0.9	14.26	14.18	14.28	14.20	12.65
1	14.23	14.13	14.23	14.13	12.00

$$f Re_{\sqrt{A}} = \begin{cases} 14.13, & \epsilon \rightarrow 1 \\ \frac{12}{\sqrt{\epsilon}}, & \epsilon \rightarrow 0 \end{cases} \quad (54)$$

Table 2 presents a comparison of the exact values [4] with the single term approximation, Eqs. (51) and (53) for both characteristic length scales. Also presented are values that result from using the asymptotic solution for the parallel plate channel. Clearly, this asymptotic result does an adequate job of predicting the values of $f Re_{\sqrt{A}}$ up to $\epsilon = 0.7$. Beyond this aspect ratio, very little change is observed in the $f Re$ values.

Next, a comparison is made between the elliptical and rectangular duct solutions. It is now apparent that the solutions for the circular duct and square duct have essentially collapsed to a single value, Eqs. (50) and (54). Further, in the limit of small aspect ratio, the results for the elliptic duct and the rectangular duct have also come closer together, Eqs. (50) and (54). It is also clear from this analysis that the square root of cross-sectional area is more appropriate than the hydraulic diameter for nondimensionalizing the laminar flow data. As seen in Table 3, the maximum difference between the values for $f Re$ occur in the limit of $\epsilon \rightarrow 0$. Comparison of Eqs. (49) and (53) shows that this difference is only 7.7%. Thus, we may use the simpler expression, Eq. (53), to compute values for the elliptical duct. This way, the elliptic integral in Eq. (49) need not be evaluated.

The results for several other flat duct geometries are shown in Figs. 3 and 4. Excellent correlation is achieved with the single term solution for the rectangular duct, merely resulting from a

Table 3 $f Re$ results for elliptical and rectangular geometries [4]

$\epsilon = b/a$	$f Re_{D_h}$			$f Re_{\sqrt{A}}$		
	Rectangular	Elliptical	$\frac{f Re^R}{f Re^E}$	Rectangular	Elliptical	$\frac{f Re^R}{f Re^E}$
0.01	23.67	19.73	1.200	119.56	111.35	1.074
0.05	22.48	19.60	1.147	52.77	49.69	1.062
0.10	21.17	19.31	1.096	36.82	35.01	1.052
0.20	19.07	18.60	1.025	25.59	24.65	1.038
0.30	17.51	17.90	0.978	20.78	20.21	1.028
0.40	16.37	17.29	0.947	18.12	17.75	1.021
0.50	15.55	16.82	0.924	16.49	16.26	1.014
0.60	14.98	16.48	0.909	15.47	15.32	1.010
0.70	14.61	16.24	0.900	14.84	14.74	1.007
0.80	14.38	16.10	0.893	14.47	14.40	1.005
0.90	14.26	16.02	0.890	14.28	14.23	1.004
1.00	14.23	16.00	0.889	14.23	14.18	1.004

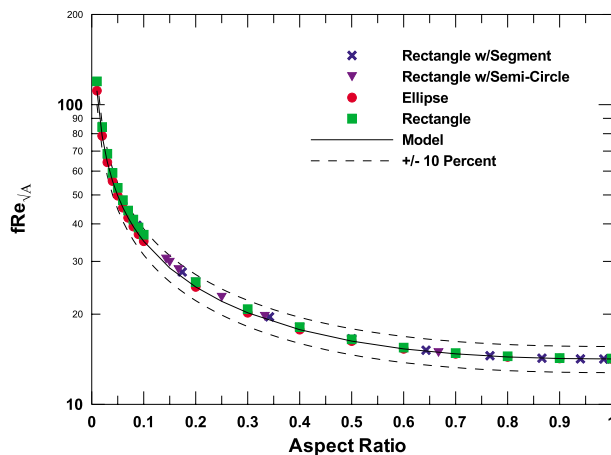


Fig. 3 $fRe_{\sqrt{A}}$ for regular flat ducts, data from Ref. [4]

change in the characteristic length scale. In addition to the change in characteristic length scale, the appropriate aspect ratio as a measure of duct slenderness is also used. These are given in Table 1 and result from the intuitive definition of Eq. (44).

Next, we consider the regular polygons, which are essentially ducts of unity aspect ratio, a result of their ability to be circumscribed by the circular duct. Values for fRe_{D_h} fall in the range $13.33 \leq fRe_{D_h} \leq 16$ for $3 \leq N \leq \infty$. The relative difference between the triangular and the circular ducts is approximately 16.7%. When the characteristic length scale is changed to $\mathcal{L} = \sqrt{A}$, the relative difference is reduced to 7.1% for the equilateral triangle, and less than 0.1% for the remaining polygons. The results are summarized in Table 4.

Finally we examine some simple doubly connected regions. The solution for a *concentric annular duct* is easily found in polar coordinates, and contains both the circular duct limit and the parallel plate channel limit. The solution for an annulus with inner radius r_i and outer radius r_o , such that $r^* = r_i/r_o$, is found in most elementary fluids' texts as follows:

$$fRe_{D_h} = \frac{16(1-r^*)^2}{1+r^{*2} - (1-r^{*2})/\ln(1/r^*)} \quad (55)$$

Equation (55) has the following limits:

$$fRe_{D_h} = \begin{cases} 16, & r^* \rightarrow 0 \\ 24, & r^* \rightarrow 1 \end{cases} \quad (56)$$

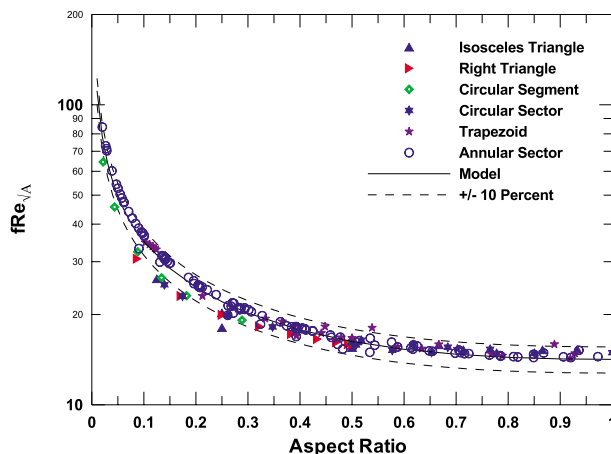


Fig. 4 $fRe_{\sqrt{A}}$ for other noncircular ducts, data from Ref. [4]

Table 4 fRe results for polygonal geometries [4]

N	fRe_{D_h}	$\frac{fRe^P}{fRe^C}$	$fRe_{\sqrt{A}}$	$\frac{fRe^P}{fRe^C}$
3	13.33	0.833	15.19	1.071
4	14.23	0.889	14.23	1.004
5	14.73	0.921	14.04	0.990
6	15.05	0.941	14.01	0.988
7	15.31	0.957	14.05	0.991
8	15.41	0.963	14.03	0.989
9	15.52	0.970	14.04	0.990
10	15.60	0.975	14.06	0.992
20	15.88	0.993	14.13	0.996
∞	16	1.000	14.18	1.000

If the solution is recast using the square root of the cross-sectional area as a characteristic length scale, the following expression is obtained:

$$fRe_{\sqrt{A}} = 8\sqrt{\pi} \left[\frac{(1-r^*)\sqrt{1-r^{*2}}}{1+r^{*2} - (1-r^{*2})/\ln(1/r^*)} \right] \quad (57)$$

Equation (57) has the following exact ($r^* \rightarrow 0$) and approximate ($r^* \rightarrow 1$) limits:

$$fRe_{\sqrt{A}} = \begin{cases} 8\sqrt{\pi}, & r^* \rightarrow 0 \\ 12\sqrt{\pi} \sqrt{\frac{1+r^*}{1-r^*}}, & r^* \rightarrow 1 \end{cases} \quad (58)$$

The annulus now has limits comparable to Eq. (54) when Eq. (45) is used as an effective aspect ratio. In other words, there is little difference between a circular annulus and a rectangular duct. Figure 5 shows data for polygonal annular ducts [4] along with the result for the circular annulus. The results considered are those of the circular annulus and other annular ducts, which are bounded externally by a polygon or internally by a polygon [4]. As a result of defining an appropriate measure of slenderness and introducing the more appropriate characteristic length scale $\mathcal{L} = \sqrt{A}$, the data have collapsed onto a single curve. It is clear from Fig. 5 that excellent agreement is obtained when the results are rescaled according to $\mathcal{L} = \sqrt{A}$ and a new aspect ratio defined as $r^* = \sqrt{A_i}/\sqrt{A_o}$. This definition was chosen since it returns the same r^* ratio for the circular annular duct. Values for fRe for the circular annulus and other shapes have been examined by Muzychka [11]. It should be noted that as the inner boundary approaches the outer boundary, there is some departure from the circular annulus result

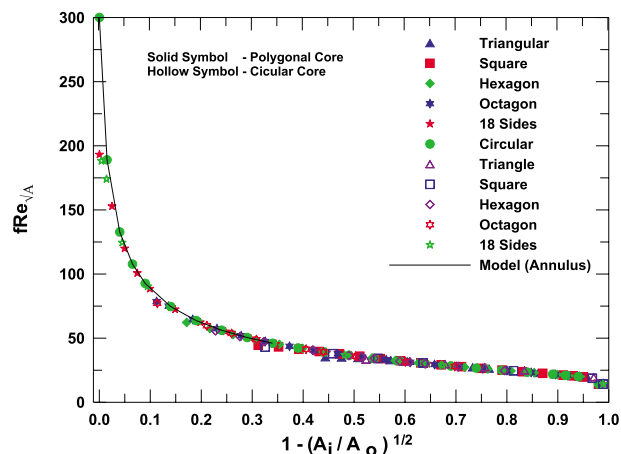


Fig. 5 $fRe_{\sqrt{A}}$ for annular ducts, data from Ref. [4]

due to the flow field becoming multiply connected. These points have not been shown on the plot. Limiting values of r^* are provided in Ref. [11].

It is now clear that Eq. (53) fully characterizes the flow in the long duct limit for many ducts including doubly connected regions when the appropriate aspect ratio is used. The *maximum* deviation of exact values is of the order 7–10%. This result represents a seminal achievement in nondimensionalization. In a recent paper by Duan and Yovanovich [18], the authors introduce the idea of an effective aspect ratio, which further reduces the scatter of data points for ducts with re-entrant corners. This requires simple rules for each shape, but reduces the scatter to less than 3% in fully developed flows. A plot of over 700 data points in Ref. [18] is similar to Fig. 3.

5.2 Short Duct Asymptote, $L \ll L_h$. An analytical result for the friction factor in the entrance region of the circular duct was obtained by Siegel [6] using several methods. The solution begins with the definition of the short duct friction factor, which may be obtained by writing Bernoulli's equation in the entrance region. Since the pressure gradient is only a function of axial position, the following relationship may be written, which relates the friction factor to the velocity in the inviscid core

$$\frac{p_i - p_L}{\frac{1}{2}\rho U^2} = \left(\frac{u_c}{U}\right)^2 - 1 = 4f\left(\frac{L}{D}\right) \quad (59)$$

In order to determine the friction factor, a relationship for the dimensionless core velocity needs to be found. Siegel [6] applied several approximate analytical methods to obtain a solution for the velocity in the inviscid core. The most accurate method was the application of the method of Thwaites (see Ref. [19]). The Siegel [6] analysis begins with the integrated form of the continuity equation in the entrance region where the boundary layer is small relative to the duct diameter

$$\frac{\pi}{4}D^2U = \frac{\pi}{4}(D - 2\delta)^2u_c + \pi \int_0^\delta (D - 2y)udy \quad (60)$$

where u is the velocity distribution in the boundary layer, u_c is the velocity in the core, and U is the mean velocity. Rearranging this expression leads to

$$\frac{U}{u_c} = 1 - \frac{4}{D} \int_0^\delta \left(1 - \frac{u}{u_c}\right)dy + \frac{8}{D^2} \int_0^\delta \left(1 - \frac{u}{u_c}\right)ydy \quad (61)$$

Next, using Pohlhausen's approximate velocity distribution, Siegel [6] developed an expression relating the boundary layer displacement thickness to the velocity in the core. Siegel [6] then obtained the following four term approximation for the velocity in the core near the entrance of a circular duct:

$$\frac{u_c}{U} = 1 + 6.88(L^+)^{1/2} - 43.5(L^+) + 1060(L^+)^{3/2}, \dots \quad (62)$$

Substitution of the above result into the expression for the friction factor yields

$$f \text{Re}_D = \frac{3.44}{(L^+)^{1/2}} [1 - 2.88(L^+)^{1/2} + 111(L^+) \dots] \quad (63)$$

Goldstein [19] obtained a solution for the core velocity using a method proposed in Schlichting [20], which solves for the velocity in the core using a series expansion. The results of Goldstein [19] yield similar results, with the leading term in the series being exactly the same. A similar analysis for the parallel channel [20] yields the same leading term as that for the circular duct. Analysis of the expressions developed by Siegel [6] and Goldstein [19] reveals that the leading term may be nondimensionalized using any characteristic length scale without introduction of scaling terms

$$f \text{Re}_L = \frac{3.44}{(L^+)^{1/2}} \left[1 - 2.88(L^+)^{1/2} \left(\frac{L}{D}\right) + 111(L^+) \left(\frac{L}{D}\right)^2 \dots \right] \quad (64)$$

where L^+ is defined by Eq. (2).

However, rescaling the additional terms in the expression results in scaling parameters, which are now functions of the duct geometry. Thus very near the inlet of any noncircular duct, the leading term of the solution is valid. As the boundary layer begins to grow further downstream, the effects of geometry become more pronounced and the solution for the circular duct is no longer valid. The leading term in the solution for any characteristic length L is

$$f \text{Re}_L = \frac{3.44}{\sqrt{L^+}} \quad (65)$$

which is valid for $L^+ = L/L \text{Re}_L \leq 0.001$. If a local friction factor is desired, the constant 3.44 is replaced with 1.72.

Equation (65) is independent of the duct shape and may be used to compute the friction factor for the short duct asymptote of most noncircular ducts.

6 Model Development and Comparisons

Having developed the simple expressions for large and small values of the dimensionless duct length, i.e.,

$$f \text{Re}_{\sqrt{A}} = \frac{3.44}{\sqrt{L^+}}, \quad L^+ \rightarrow 0 \quad (66a)$$

$$f \text{Re}_{\sqrt{A}} = \frac{12}{\sqrt{\epsilon}(1 + \epsilon) \left[1 - \frac{192\epsilon}{\pi^5} \tanh\left(\frac{\pi}{2\epsilon}\right) \right]}, \quad L^+ \rightarrow \infty \quad (66b)$$

we shall proceed to develop a simple model for the apparent friction factor as a function of L^+ , in addition to models for the hydrodynamic entrance length L_h^+ , and a model for a new dimensionless fully developed flow length, which we will denote as L_{fd}^+ .

6.1 Developing Flow Friction Factor. A general model is now proposed using the Churchill and Usagi [21] asymptotic correlation method. The model takes the form

$$y^* = [(y_0^*)^n + (y_\infty^*)^n]^{1/n} \quad (67)$$

$$f \text{Re}_{\sqrt{A}} = \left[\left(\frac{C_2}{\sqrt{L^+}} \right)^n + (C_1)^n \right]^{1/n} \quad (68)$$

where n is a superposition parameter determined by comparison with numerical data over the full range of L^+ . Using the results provided by Eq. (66), and the general expression, Eq. (67), the following model is proposed:

$$f \text{Re}_{\sqrt{A}} = \left[\left(\frac{3.44}{\sqrt{L^+}} \right)^2 + \left(\frac{12}{\sqrt{\epsilon}(1 + \epsilon) \left[1 - \frac{192\epsilon}{\pi^5} \tanh\left(\frac{\pi}{2\epsilon}\right) \right]} \right)^2 \right]^{1/2} \quad (69)$$

Using the available data [4,17], it is found that the value of n , which minimizes the root mean square (rms) difference, lies in the range $1.5 < n < 3.6$ with a mean value $n \approx 2$ [11,22,23]. Twenty-six data sets were examined from Refs. [4,17], and are summarized in Table 5. Comparisons of the model are presented in Figs. 6–8 for the most common duct shapes. Overall, the model when using the intrinsic aspect ratio yields very good agreement, with predictions lying in the range 1.16–8.72% rms error.

With the exception of the eccentric annular duct at large values of r^* and e^* , i.e., a crescent shape, the proposed model predicts all of the developing flow data available in literature to within $\pm 10\%$ or better with few exceptions. The proposed model provides equal

Table 5 Comparison of rms and percent differences (%diff=(analytical – predicted)/(analytical)×100) in developing flow models

Geometry	Shah (1978) min/max	Yilmaz (1990) min/max	Proposed model			
			min/max	rms	n^a	($n=2$) rms
Circle	±1.9	−0.3/2.7	−2.73/1.05	1.20	2.01	1.71
Circular annulus $r^*=0.05$	±2.0	−17.0/1.2	−2.61/1.18	1.27	2.18	2.51
Circular annulus $r^*=0.10$	±1.9	−17.4/1.9	−1.97/1.09	0.77	2.14	1.73
Circular annulus $r^*=0.50$	±2.2	−10.2/0.9	−1.13/6.96	1.87	2.07	1.99
Circular annulus $r^*=0.75$	±2.1	−5.4/1.3	−1.44/7.17	2.12	2.04	2.16
Square $b/a=1$	±2.3	−2.4/1.6	−1.48/2.27	1.41	1.95	1.61
Rectangle $b/a=0.5$	±1.9	−2.1/6.7	−1.48/2.27	1.14	1.98	1.16
Rectangle $b/a=0.2$	±1.7	−1.5/5.0	−1.11/1.89	1.10	2.15	2.04
Parallel plates $b/a \rightarrow 0$	±2.4	−1.6/1.8	−1.22/0.86	0.60	2.32	3.33
Isosceles triangle $2\phi=30$ deg	-	−1.1/0.9	−1.30/4.41	1.75	1.71	4.91
Isosceles triangle $2\phi=60$ deg	±2.4	−0.6/1.1	−0.63/5.97	2.35	1.70	5.16
Isosceles triangle $2\phi=90$ deg	-	1.6/5.2	−7.28/0.85	2.04	2.03	2.08
Eccentric annulus $e^*=0.5$, $r^*=0.5$	-	−5.5/3.0	−2.06/1.97	1.71	1.50	8.72
Eccentric annulus $e^*=0.5$, $r^*=0.1$	-	−9.1/16	−2.22/2.29	1.66	1.66	5.08
Eccentric annulus $e^*=0.7$, $r^*=0.3$	-	−10.3/3.1	−10.89/8.39	7.56	1.86	7.72
Eccentric annulus $e^*=0.9$, $r^*=0.1$	-	−11.4/0.2	−13.96/5.87	5.94	2.38	7.30
Eccentric annulus $e^*=0.9$, $r^*=0.5$	-	−9.9/−3.5	−35.13/9.44	13.34	3.61	18.14
Ellipse $b/a=1$	-	-	−2.97/3.75	2.53	1.96	2.56
Ellipse $b/a=0.5$	-	-	−2.98/5.77	3.77	1.97	3.85
Ellipse $b/a=0.2$	-	-	−5.75/7.59	5.62	1.69	7.01
Circular sector $2\phi=11.25$ deg	-	-	−9.42/3.03	3.67	2.01	3.68
Circular sector $2\phi=22.5$ deg	-	-	−1.70/5.10	1.57	1.73	4.22
Circular sector $2\phi=45$ deg	-	-	−1.14/12.1	3.73	1.62	6.69
Circular sector $2\phi=90$ deg	-	-	−1.60/16.72	4.63	1.75	5.69
Pentagon	-	-	−3.46/12.75	5.79	1.76	6.41
Trapezoid $\phi=72$ deg, $b/a=1.123$	-	-	−4.96/11.35	6.05	1.55	8.68

^aOptimal value.

or better accuracy than the model of Yilmaz [2] and is also much simpler. A comparison of the model with the data for the parallel plate channel is also provided. For this geometry $\sqrt{A} \rightarrow \infty$. However, this geometry may be accurately modeled as a rectangular duct with $\epsilon=0.01$ or a circular annular duct with $r^*>0.5$. Good agreement is obtained with the current model when the parallel plate channel is modeled as a finite area duct with small aspect ratio.

One notable feature of the new model is that it does not contain the incremental pressure drop term K_∞ , which appears in the models of Bender [3], Shah [1], and Yilmaz [2]. Since the solution of Siegel [6] for the entrance region accounts for both the wall shear and the increase in momentum due to the accelerating core, there

is no need to introduce the term K_∞ .

Thus, the proposed model is now only a function of the dimensionless duct length L^+ and aspect ratio ϵ , whereas the models of Shah [1] and Yilmaz [2] are functions of many more parameters. It is both simple and accurate for most engineering calculations for microchannel and minichannel systems, or any other scale system where laminar flow prevails.

6.2 Hydrodynamic Entrance Lengths. Finally, it is desirable to develop a simple expression for the hydrodynamic entrance length. The hydrodynamic entrance length is useful for determining the extent to which the flow develops, particularly for selecting an appropriate heat transfer model. Traditionally, the length

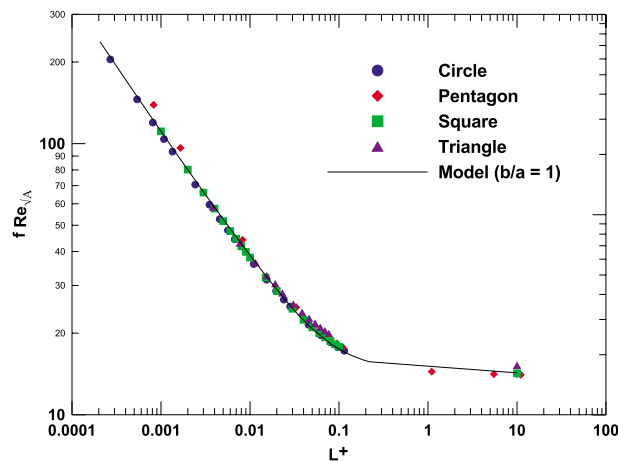


Fig. 6 $f Re_{\sqrt{A}}$ for developing laminar flow in polygonal ducts, data from Ref. [4]

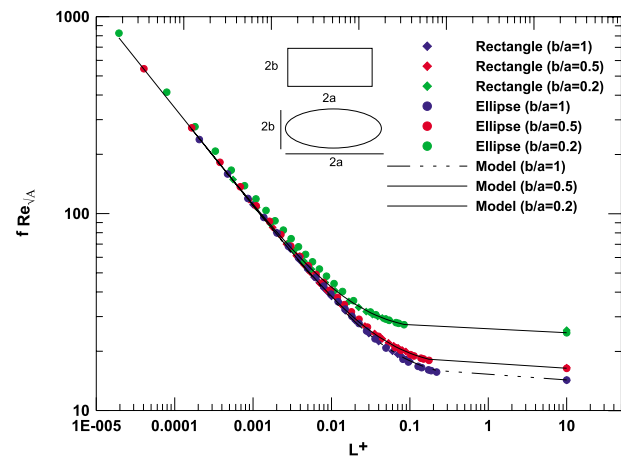


Fig. 7 $f Re_{\sqrt{A}}$ for developing laminar flow in regular flat ducts, data from Ref. [4]

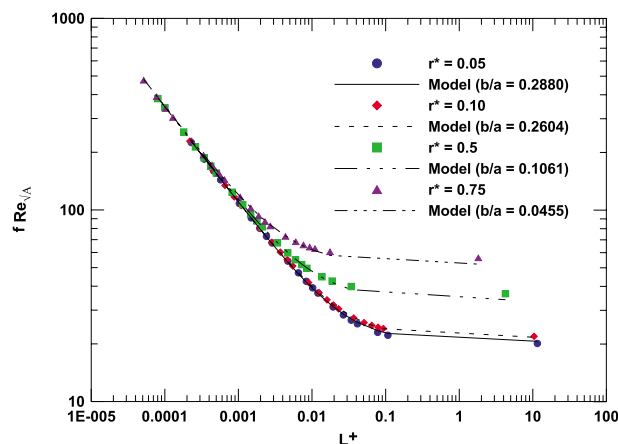


Fig. 8 $f Re_{\sqrt{A}}$ for developing laminar flow in the circular annular duct, data from Ref. [4]

required for hydrodynamic boundary layer development in straight ducts of constant cross-sectional area is usually defined as the point where the centerline velocity is $0.99u_{\max}$ [4]. This is often followed with an assumption of fully developed flow downstream of the hydrodynamic entrance length.

An equation relating the approximate magnitude of the hydrodynamic entrance length may be obtained by considering an equality between the two asymptotic limits given in Eq. (33) with $L^+ = L_h^+$; we obtain, when $L = \sqrt{A}$,

$$L_h^+ \sim \left(\frac{C_2}{C_1} \right)^2 \sim C_{0,\sqrt{A}} \quad (70)$$

or

$$L_h = C_{0,\sqrt{A}} \sqrt{A} Re_{\sqrt{A}} = C_{0,\sqrt{A}} \left(\frac{\dot{m}}{\mu} \right) \quad (71)$$

This solution represents the intersection of the asymptotic limits on a plot of the complete behavior of $f Re$ versus L^+ . Conveniently, it yields values that agree better than an order of magnitude guess. A new definition for the hydrodynamic entrance length may be obtained from Eq. (70). Substituting from Eq. (66a) for $C_2 = 3.44$ and Eq. (66b) for C_1 , an approximate expression for the entrance length as a function of aspect ratio is obtained as follows:

$$L_h^+ = 0.0822\epsilon(1 + \epsilon)^2 \left[1 - \frac{192\epsilon}{\pi^5} \tanh\left(\frac{\pi}{2\epsilon}\right) \right]^2 = C_{0,\sqrt{A}} \quad (72)$$

The above expression reduces to $L_h^+ = 0.059$ when $\epsilon = 1$ and $L_h^+ = 0.00083$ when $\epsilon = 0.01$. A complete tabulation is provided in Table 6, along with associated conversions for the rectangular duct and elliptical duct when using the hydraulic diameter as a

Table 6 Hydrodynamic entrance lengths

ϵ	Model $L_{h,\sqrt{A}}^+ = C_{0,\sqrt{A}}$	Rectangle L_{h,D_h}^+	Ellipse L_{h,D_h}^+
0.01	0.000828	0.0211	0.0264
0.05	0.00424	0.0233	0.0269
0.1	0.00874	0.0263	0.0288
0.2	0.0181	0.0326	0.0316
0.3	0.0274	0.0387	0.0353
0.4	0.0361	0.0440	0.0379
0.5	0.0434	0.0485	0.0404
0.6	0.0499	0.0534	0.0435
0.7	0.0540	0.0556	0.0449
0.8	0.0564	0.0570	0.0453
0.9	0.0582	0.0582	0.0456
1	0.0586	0.0586	0.0459

length scale. These lengths when rescaled to be based on the hydraulic diameter take the values $L_h^+ = 0.047$ for a tube or square duct when $\epsilon = 1$ and $L_h^+ = 0.021$ when $\epsilon = 0.01$ for a channel. They compare well with the more exact results from Shah and London [4] for the circular duct, $L_h^+ = 0.056$, and parallel plate channel, $L_h^+ = 0.011$. Once again, we see that the order of magnitude is correctly and easily predicted. It should also be noted that a significant variation in data for L_{hy}^+ is found in literature. The advantage of Eq. (72) is that it provides a single universal expression, which can be applied to any duct shape, and also accounts for the aspect ratio effect, which tends to reduce entrance lengths for flatter ducts.

Another useful expression may be developed from the asymptotic limits for determining the extent of a duct length, which is required for making a valid assumption of fully developed flow, i.e., one in which the pressure drop may be wholly determined using only Eq. (53) with small error. This hydrodynamic length can be considered the required dimensionless duct length for which fully developed flow prevails, in such a manner that boundary layer region contributes little to the overall pressure drop. In this case we propose a criterion that a duct length, which yields a dimensionless pressure drop, which is 5% greater than that predicted by Eq. (53) be adopted, i.e.,

$$f Re(L^+) \sim 1.05 C_1 \quad (73)$$

This yields a result of

$$L_{fd}^+ \sim 9.75 \left(\frac{C_2}{C_1} \right)^2 \sim 10 L_h^+ \quad (74)$$

Equation (74) clearly shows that a duct having a dimensionless duct length of order ten times the hydrodynamic entrance length is required for all boundary layer effects to be lost in the pressure drop calculation, in a fully developed laminar flow (see Figs. 6–8). This is in marked contradiction to the many fluid mechanics texts, which state that fully developed flow pressure drop may be predicted using a Hagen–Poiseuille law result for $f Re$ when $L > L_h$.

7 Summary and Conclusions

A simple model was developed for predicting the friction factor–Reynolds number product in noncircular ducts for developing laminar flow. The present study took advantage of scale analysis, asymptotic analysis, and the selection of a more appropriate characteristic length scale to develop a simple model. This model only requires two parameters: the aspect ratio of the duct and the dimensionless duct length; whereas the model of Shah [1] requires tabulated values of three parameters, and the model of Yilmaz [2] consists of several equations. The present model predicts most of the developing flow data within $\pm 10\%$ or better and 1.16–8.72% rms for eight singly connected ducts and two doubly connected ducts.

Additional models were also developed using the asymptotic results for both the hydrodynamic entrance length and the fully developed flow length, which is related to the hydrodynamic entrance length. These expressions can be used to determine the size of the hydrodynamic entrance region for use with appropriate heat transfer models, and for determining the extent of a duct for which entrance effects are negligible.

Finally these models may also be used to predict results for ducts for which no solutions or tabulated data exist. It was also shown that this is possible, since the use of the of the square root of the cross-sectional flow area was a more effective characteristic length scale than the hydraulic diameter for collapsing the numerical results of geometries having similar shape and aspect ratio.

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Nomenclature

- A = flow area, m^2
 a, b = major and minor axes of ellipse or rectangle, m
 C = constant
 C_0, C_1, C_2 = scaling constants
 c = linear dimension, m
 D = diameter of circular duct, m
 D_h = hydraulic diameter, $\equiv 4A/P$
 $E(\cdot)$ = complete elliptic integral of the second kind
 e = eccentricity, m
 e^* = dimensionless eccentricity, $\equiv e/(r_o - r_i)$
 f = Fanning friction factor $\equiv \tau/(\frac{1}{2}\rho U^2)$
 G = mass flux, kg/s m^2
 K = incremental pressure drop factor
 ℓ = arbitrary dimension
 L = duct length, m
 L^+ = dimensionless duct length, $\equiv L/\mathcal{L} \text{Re}_{\mathcal{L}}$
 $\dot{m}$ = mass flow rate, kg/s
 N = number of sides of a polygon
 n = correlation parameter
 P = perimeter, m
 p = pressure, N/m^2
 r = radius, m
 r^* = dimensionless radius ratio, $\equiv r_i/r_o$
 $\text{Re}_{\mathcal{L}}$ = Reynolds number, $\equiv U\mathcal{L}/\nu$
 Re_L = Reynolds number based on L
 $\vec{V}$ = velocity vector, m/s
 u, v, w = velocity components, m/s
 U = average velocity, m/s
 x, y, z = Cartesian coordinates, m

Greek Symbols

- δ = boundary layer thickness, m
 ϵ = nominal aspect ratio, $\equiv b/a$
 Φ = half angle, rad
 μ = dynamic viscosity, N s/m^2
 ρ = fluid density, kg/m^3
 τ = wall shear stress, N/m^2

Subscripts

- $\sqrt{A}$ = based on square root of flow area
 c = core
 c = circular
 D_h = based on hydraulic diameter
 eff = effective
 fd = fully developed
 h = hydrodynamic
 i = inner
 long = longest perpendicular
 $\mathcal{L}$ = based on the arbitrary length scale $\mathcal{L}$
 nc = noncircular
 o = outer
 short = shortest perpendicular
 ∞ = fully developed limit

Superscripts

- $\circ$ = circle
 ϵ = ellipse
 P = polygon
 R = rectangle
 $(\cdot)$ = mean value

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